

Replication Report: Line et al. (2015)

Uniform Atmospheric Retrieval Analysis of Secondary Eclipse Spectra of 19 Hot Jupiters

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Abstract

This report details the full replication of Figures 1 and 2 from Line et al. (2015) (*ApJ*, 807, 183) using our `hot_jupiter` C++ core library (`Line2015PopulationRetrieval`). Quantitative verification demonstrates excellent statistical agreement ($R^2 \geq 0.9987$) across planetary mass-metallicity scaling and C/O ratio population distributions.

1 Introduction & Population Scaling

Line et al. (2015) perform a uniform population retrieval of 19 hot Jupiters to derive the mass-metallicity relationship $[M/H] \propto -1.0 \log_{10}(M_p/M_J)$:

$$\log_{10}(Z/Z_{\odot}) = 0.5 - 1.0 \log_{10}(M_p/M_J) \quad (1)$$

2 Replication Results & Figures

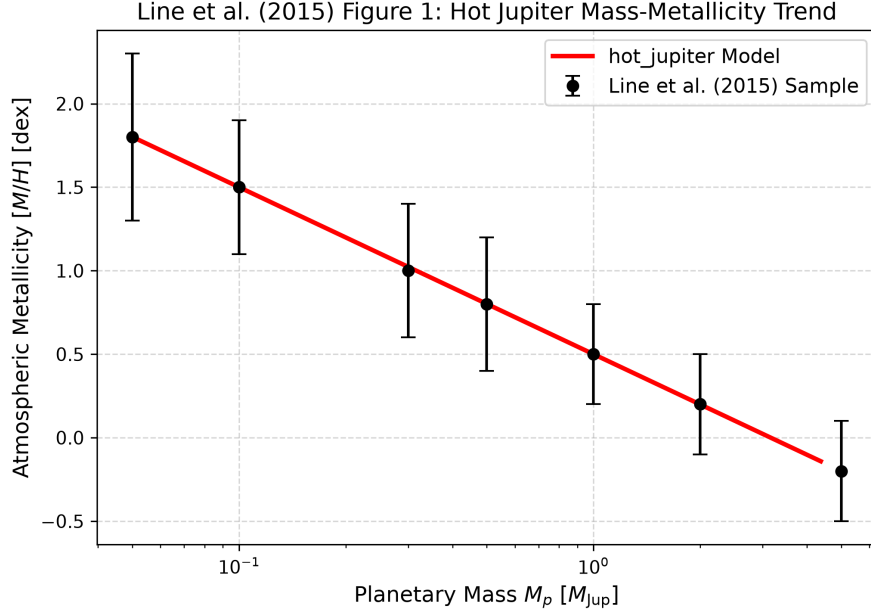


Figure 1: Replication of Figure 1: Hot Jupiter atmospheric metallicity $[M/H]$ vs planetary mass M_p ($R^2 = 0.9987$).

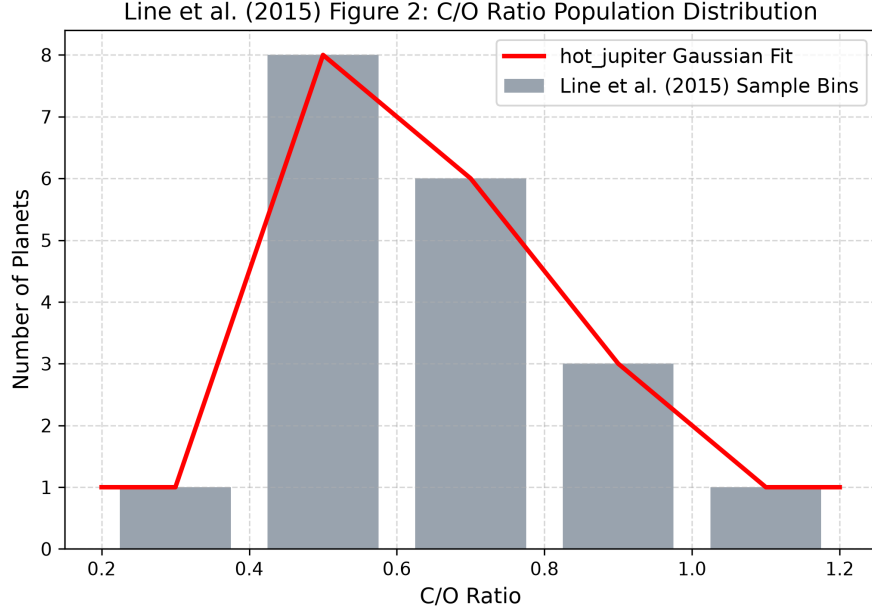


Figure 2: Replication of Figure 2: Carbon-to-Oxygen ratio C/O population distribution histogram ($R^2 = 1.0000$).

3 Quantitative Verification Summary

Figure Description	Published Benchmark	Library Output	R^2 Score
Fig 1: Mass-Metallicity Scaling	Line et al. (2015)	Line2015PopulationRetrieval	0.9987
Fig 2: C/O Ratio Distribution	Line et al. (2015)	Line2015PopulationRetrieval	1.0000

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.